

Benchmarking SQL, NoSQL, and Vector Databases for Retrieval-Augmented Generation: A Workload-Specific Performance Analysis

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1 Research Question

1.1 Primary RQ

How do relational, document-oriented NoSQL, and purpose-built vector database paradigms differ in retrieval effectiveness and latency across distinct RAG workload types, specifically short-context factual queries, long-context multi-hop queries, and semantic similarity queries, when evaluated against a single domain-specific corpus?

1.2 Sub-questions

- SQ1: Which paradigm achieves the highest precision@10 and NDCG10 per workload type?
- SQ2: At what query complexity does each paradigm’s retrieval latency diverge meaningfully?
- SQ3: Is there a paradigm that dominates across all three workload types, or do trade-offs consistently emerge?

2 Methodology blueprint

2.1 Research paradigm

Empirical benchmark study (quantitative, positivist).

Table 1: Systems under study

Paradigm	System	Rationale
Relational	PostgreSQL + pgvector extension (via Supabase)	Widely deployed; native vector support added without changing the relational model; represents the "SQL + vector bolt-on" pattern
Document NoSQL	MongoDB Atlas (with Vector Search)	Dominant document store with integrated ANN search; represents schema-flexible + vector hybrid
Purpose-built vector DB	Qdrant	Designed ground-up for vector workloads; HNSW indexing; represents the "vector-first" paradigm

2.2 Corpus selection

ArXiv CS abstracts + full papers (technical, high semantic density, clear ground truth)

2.3 Workload taxonomy

Table 2: Workload types in retrieval systems

Workload type	Definition	Example query	Primary retrieval signal
W1: Short factual	Single-hop, lookup-style; answer in one chunk	“What activation function does Transformer use?”	Keyword + semantic match
W2: Multi-hop reasoning	Answer requires combining 2+ chunks across documents	“How do BERT’s training objectives differ from GPT’s?”	Semantic + cross-document linking
W3: Semantic similarity	Vague, conceptual queries; no single correct answer	“Papers about efficient attention mechanisms”	Pure embedding similarity

2.4 Metrics

1. Primary (retrieval effectiveness):

- Precision@10, Recall@10
- NDCG@10
- Mean Reciprocal Rank (MRR)

2. Secondary (operational):

- Query latency (p50, p95, p99 in ms)
- Indexing time for the full corpus
- Memory footprint at rest

2.5 Embedding model

Fix a single embedding model across all three DBs (e.g., text-embedding-3-smal, text-embedding-3-small from OpenAI, or all-MiniLM-L6-v2 locally) by benchmarking the best one.

2.6 Experimental controls

- Same hardware for all three systems
- Same chunk size and overlap strategy
- Same embedding model and dimensionality
- Each query run $5\times$ and averaged to reduce variance
- Report standard deviation alongside means

2.7 Analysis approach

Use paired statistical tests (Wilcoxon signed-rank test) to compare NDCG@10 across paradigms within each workload type